

Maksim Kosmakov

Curriculum Vitae

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Academic Appointments

- 2026–present **Visiting Assistant Professor**, *University of Toledo*
Department of Mathematics and Statistics; since August 2026
- 2024–2026 **Visiting Assistant Professor**, *University of Cincinnati*
Department of Mathematical Sciences
- 2023–2024 **Visiting Assistant Professor**, *Gettysburg College*
Department of Mathematical Sciences

Education

- 2018–2023 **Ph.D. in Mathematics**, *Indiana University – Purdue University Indianapolis*
Department of Mathematical Sciences
Advisor: V. Tarasov
- 2015–2017 **Master of Physics (Diploma with distinction)**, *St. Petersburg State University*
Faculty of Physics, Department of Mathematics and Mathematical Physics
- 2011–2015 **Bachelor of Physics (Diploma with distinction)**, *St. Petersburg State University*
Faculty of Physics, Department of Mathematics and Mathematical Physics
Title: *One-loop divergences for the effective action in the Weinberg–Salam model*
Advisor: Dr. L. D. Faddeev

PhD Thesis

- Title *Certain aspects of quantum and classical integrable systems*
- Supervisors V. Tarasov & A. Its
- Description In this thesis we derive new combinatorial formulas for vector-valued weight functions for the evaluation modules over the Yangians $Y(\mathfrak{gl}_n)$. We obtain them using the nested algebraic Bethe ansatz method. We also describe the asymptotic behavior of the radial solutions of the negative tt^* -equation via the Riemann–Hilbert problem and the Deift–Zhou nonlinear steepest descent method.

Master's Thesis

- Title *Mathematical problems of quantum field theory*
- Supervisors L. D. Faddeev & A. A. Fedotov
- Description This thesis explored the idea of non-Fock representations of the canonical commutation relations in QED.

Publications and Preprints

- 2026 Maksim Kosmakov. *A Two-HCIZ Gaussian Matrix Model for Non-Intersecting Brownian Bridges*. Journal of Statistical Physics 193, 93.
- 2026 Oleg Asipchuk, Maksim Kosmakov, Pavel Zatitskii. *Rigidity in the planar Ulam floating body problem with perimetral densities $\sigma = 1/8, 3/8$ under central symmetry*. arXiv:2607.08338.
- 2026 Roozbeh Gharakhloo, Maksim Kosmakov, Kenta Miyahara. *Reduction of Multiple Orthogonal Polynomials to Standard Orthogonal Polynomials*. arXiv:2606.27594.
- 2026 Oleg Asipchuk, Maksim Kosmakov, Pavel Zatitskii. *Rigidity in the Planar Ulam Floating Body Problem with perimetral density $\sigma = 1/6$* . International Mathematics Research Notices 2026 (18), rnag203 · arXiv:2604.10330.
- 2025 Martin A. Guest, Alexander R. Its, Maksim Kosmakov, Kenta Miyahara, Ryosuke Odoi. *Connection formulae for the radial Toda equations II*. Nonlinearity 38, 105016.
- 2025 Maksim Kosmakov, Vitaly Tarasov. *New combinatorial formulae for nested Bethe vectors*. SIGMA 21, 060.
- 2025 Martin A. Guest, Alexander R. Its, Maksim Kosmakov, Kenta Miyahara, Ryosuke Odoi. *Connection formulae for the radial Toda equations I*. Nonlinearity 38, 035015.
- 2025 Maksim Kosmakov, Vitaly Tarasov. *New combinatorial formulae for nested Bethe vectors II*. Letters in Mathematical Physics 115, 12.
- 2026 Maksim Kosmakov. *Factor Neutralization and Risk-Budgeted Scaling: Evidence from Long-Short Anomaly Portfolios*. SSRN.
- 2026 Maksim Kosmakov. *Capital-Level Factor Neutrality with Projected Partial Rebalancing*. SSRN.
- 2023 Kosmakov, M. *Certain Aspects of Quantum and Classical Integrable Systems*. Ph.D. dissertation, Purdue University, Indianapolis.

Conferences and Talks

- 2026 Synergies between Geometry, Probability, and Computation in High Dimensions. ICERM, Brown University · Providence, Rhode Island August 1 · Lightning talk: “Gamma Anticoncentration and Matrix Slicing: Work in Progress.”
- 2026 SIAM Annual Meeting (AN26). July 6–10 · Huntington Convention Center, Cleveland, Ohio July 7 · Talk: “Matrix Model for Mixed Multiple Orthogonal Polynomials.”
- 2026 Ohio River Analysis Meeting. March 21–22 · University of Kentucky, Lexington
- 2025 Midwestern Workshop on Asymptotic Analysis. October 10–12 · Purdue University Fort Wayne
- 2025 Waves 2025. April 14–16 · Athens, Georgia. Thirteenth International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory.
- 2024 Riemann–Hilbert problems, Toeplitz matrices, and applications. March 4–8 · American Institute of Mathematics, Pasadena
- 2023 Midwestern Workshop on Asymptotic Analysis. October 13–15 · IUPUI, Indianapolis
- 2016 Science and Progress. International Student Conference
- 2015 Science and Progress. International Student Conference

Honors and Awards

- 2023 Charalambos D. Aliprantis Prize
- 2022 Outstanding Advanced Graduate Student in Mathematics Award (IUPUI)
- 2021 Yuri Abramovich Memorial Scholarship
- 2015 Diploma Grade 2, Student Olympiad in Mathematics (ITMO)
- 2013 Special Scholarship of the Government of St. Petersburg in Mathematics and Physics
- 2012 Diploma Grade 3, Student Olympiad in Mathematics (ITMO)

Departmental Leadership

- 2022–2023 AMS Graduate Student Chapter Vice President
- 2021–2022 AMS Graduate Student Chapter President
- 2019–2021 AMS Graduate Student Chapter Vice President

Computer Skills

- Languages Python
- Software Maple, MATLAB

In Progress

Kosmakov, M. *Dressing procedure for the singular Riemann–Hilbert problem.*

Guest, M. A., Its, A. R., Kosmakov, M., and Miyahara, K. *Real forms of the affine Toda equation of A_2 type and asymptotics of smooth-near-zero solutions.*

Houdré, C., and Kosmakov, M. *Longest common increasing subsequences for random words and their asymptotics.*

Teaching Experience

Courses at University of Cincinnati

- Fall 2025 **Math 1062**, *Calculus II*, University of Cincinnati
- Fall 2025, **Math 2073**, *Differential Equations*, University of Cincinnati
- Summer 2025
- Fall 2024, **Math 1026**, *Pre-Calculus*, University of Cincinnati
- Spring 2025
- Fall 2024 **Math 1061**, *Calculus I*, University of Cincinnati
- Fall 2024 **Math 1006**, *Math Reasoning*, University of Cincinnati

Courses at Gettysburg College

- Spring 2024 **Math 221**, *Multivariable Calculus*, Gettysburg College
- Fall 2023 **Math 112**, *Calculus II*, Gettysburg College
- Fall 2023 **Math 225**, *Differential Equations*, Gettysburg College

Courses at IUPUI

- Spring 2023 **Math 111**, *Algebra*, IUPUI
- Spring 2023 **Math 166**, *Analytic Geometry and Calculus II*, IUPUI
- Fall 2022 **Math 171**, *Multidimensional Mathematics*, IUPUI
- Fall 2022 **Math 222**, *Calculus for Technology II*, IUPUI
- Fall 2021 **Math 266**, *Ordinary Differential Equations*, IUPUI
- Fall 2021 **Math 110**, *Fundamentals of Algebra*, IUPUI
- Fall 2020 **Math 221**, *Calculus for Technology I*, IUPUI
- Fall 2020 **Math 165**, *Analytic Geometry and Calculus I*, IUPUI
- Summer 2020 **Math 166**, *Brief Survey of Calculus*, IUPUI
- Spring 2020 **Math 166**, *Analytic Geometry and Calculus II*, IUPUI
- Spring 2020 **Math 118**, *Finite Mathematics*, IUPUI
- Fall 2019 **Math 118**, *Finite Mathematics*, IUPUI

Other

- 2018–2020 **Mathematics Tutor**, *Mathematics Assistance Center*, IUPUI
- 2015–2017 **Teacher of Physics**, *School 533 (YMS)*, Saint Petersburg